[411]

A Memoir on the Theory of Reciprocal Surfaces. (continued)

Article 1 (60.)**.** in my Memoir on Cubic Surfaces was obtained without the aid of the formula now in question) I endeavour by means of the contained results to find the values of the unknown coefficients h, g, χ, λ, μ, ν.

For a cubic surface $n = 3$, and in fact for all the cases except the cubic scrolls (XXII and XXIII), the formula is

$$\theta' = 54 - hC - gB - \lambda' j' - \mu'\chi' - \nu'\theta' - h'C' - g'B',$$

and applying this to the several cases of cubic surfaces as grouped together in the Table, and referred to by the affixed roman numbers, the resulting equations are

$$\begin{aligned}
54 &= 54, && \text{(I)}\\
30 &= 54 - f, && \text{(II)}\\
18 &= 54 - g - 16\nu', && \text{(III)}\\
13 &= 54 - 2h - \lambda', && \text{(IV)}\\
6 &= 54 - h - g - \mu' - 3\nu', && \text{(VI)}\\
3 &= 54 - 3h - 3\lambda', && \text{(VIII)}\\
1 &= 54 - 2g - 16\nu' - g', && \text{(IX)}\\
0 &= 54 - 4h - 6\lambda', && \text{(XIII)}\\
0 &= 54 - h - 2g - \mu' - g', && \text{(XVI)}\\
0 &= 54 - 3g - 3g', && \text{(XVII)}\\
0 &= 54 - 2h - g - \lambda' - 2\mu', && \text{(XXI)}
\end{aligned}$$

which are all satisfied if only

$$h = 24, \quad g = 9, \quad g + 16\mu' = 36, \quad g + 2\mu' = 12, \quad g + g' = 18, \quad \lambda' = -7.$$

Article 2 (61.)**.** If we apply to the same surfaces the reciprocal equation (observing that the Table of Singularities for the reciprocal surfaces, which is given by interchanging the upper and lower halves of the original Table of Singularities, is as follows):

$$\begin{aligned}
0 &= 54432 - 54432, && \text{(I)}\\
0 &= 27851 - 27846 - \lambda', && \text{(II)}\\
0 &= 18180 - 18318 - g' - 16\nu, && \text{(III)}\\
0 &= 11765 - 11756 - 2\mu' - \lambda, && \text{(IV)}\\
0 &= 6917 - 6584 - h' - g' - \mu - 3\nu, && \text{(VI)}\\
0 &= 3534 - 3522 - 3\lambda, && \text{(VIII)}\\
0 &= 3024 - 3144 - 2g' - 16\nu - 5\nu, && \text{(IX)}\\
0 &= 1433 - 1186 - 2h' - g' - \lambda - 2\mu, && \text{(XIII)}\\
0 &= 518 - 504 - 4h' - 6\lambda, && \text{(XVI)}\\
0 &= 383 - 322 - h' - 2g' - g + 2\mu, && \text{(XVII)}\\
0 &= 54 - 54, && \text{(XXI)}
\end{aligned}$$

viz. these are

$$3h' + 3\lambda = 9, \tag{II}$$
$$g' + 16\nu + g = 45, \tag{III}$$
$$2\mu' + \lambda + 2\mu = 47, \tag{IV}$$
$$h' + g' + \mu + 3\nu = 12, \tag{VI}$$
$$3\lambda = 8,\ \text{(error, should be)}\ 3\lambda = 12, \tag{VIII}$$
$$2g' + 16\nu + 5\nu = -120, \tag{IX}$$
$$2h' + g' + \lambda + 2\mu = 14, \tag{XIII}$$
$$4h' + 6\lambda = 14, \tag{XVI}$$
$$h' + 2g' + g + 2\mu = 61, \tag{XVII}$$
$$2h' + 3g + 3g' = 54, \tag{XXI}$$

all satisfied if only

$$h' = 5, \quad g' + 16\nu = -138, \quad h' + 2\mu = 38, \quad g + g' = 18, \quad \lambda = -1.$$

Article 3 (62.)**.** I remark however that the cubic scroll XXII or XXIII gives $\theta' = 54 - (330 - 272) - 2(\lambda + \lambda')$, that is, $\lambda + \lambda' = -2$, instead of $\lambda + \lambda' = -8$. Whence if $U = 0$ be the equation of a scroll, the equation $UV = 0$, containing U as a factor, has for every point of the section constituting the flecnodal curve; but the numerical investigation is in fact not applicable to a scroll, and I am nevertheless surprised that the surface has really a definite flecnodal curve.

Article 4 (63.)**.** Combining the two sets of results, we find

$$\begin{aligned} h &= 24, & h' &= 5, \\ g &= 9, & g' &= 10 + \sigma, \\ \lambda &= \lambda', & \lambda' &= -7, \\ \mu &= \mu', & \mu' &= 18 - \tfrac{1}{2}\sigma, \\ \nu &= \nu', & \nu' &= 1 - \tfrac{1}{16}\sigma, \end{aligned}$$

and the formula thus is

$$\begin{aligned} \beta' = 2n(n-2)(11n-24) - (110n - 272)b + 44g - (116n - 303)c + 4h \\ + \sigma\{19g + 248\tau + 198\chi - 24C + j - 10\chi + \theta\} \\ - 5C - 18\Gamma - 6\chi' - \lambda j - \lambda' j' + \tfrac{1}{2}g(-16B - 8\chi - \theta + 16B' - 8\chi' - \theta'). \end{aligned}$$

Article 5 (64.)**.** where χ, χ', g are constants which remain to be determined. The cubic surfaces fail to determine them, for the reason that in all of them we have $i = 0$, $i' = 0$; and

$$165 + 8\chi + \theta = 16g + 8\chi' + \theta',$$

of which I do not perceive any contradiction. Substituting herein for g, χ, θ their last values, this is

$$\begin{aligned} \beta' = 2n(n-2)(11n-24) + b(-66n + 184) + c(-93n + 240) \\ + 141\beta + \tfrac{51}{4}\gamma + 66\iota \\ - \chi' i' + 7j' - 6\chi' - \tfrac{9}{4}\theta' - 5C' - 18B' \\ - (\chi + 87)i - 21j - \tfrac{41}{2}\chi + \tfrac{51}{2}\theta - 24C \\ + \tfrac{1}{16}g(-16B - 8\chi - \theta + 16B' + 8\chi' + \theta'). \end{aligned}$$

Article 6 (65–68.)**. Article Nos. 65 to 68.**

Article 7 (65.). a, class of curve of intersection by any plane.
δ, number of double tangents of curve of intersection.
κ, number of its inflexions.
n', class of the surface.
α', class of curve of intersection by any plane.
δ', number of double tangents of curve of intersection.
κ', number of its inflexions.
V, class of node-couple torse.
k', number of its apparent double planes.
i', number of its triple planes.
q, its order.
p, order of node-couple curve.
y, number of pinch-planes.
c', class of spinode torse.
A', number of its apparent double planes.
r', its order.
σ', order of spinode curve.
g', number of off-planes.
γ', number of close-planes.
θ', number of common planes of node-couple and spinode torses.
i', number of common planes, stationary planes of the spinode torse.
j', number of common planes, stationary planes of node-couple torse.
χ', number of common planes, not stationary planes of either torse.
B, number of bitropes of surface.
C, number of its cnictropes.

Article 8 (66.). ρ, number of intersections, stationary points on nodal curve.
ι, number of intersections, stationary points on cuspidal curve.
i, number of intersections, not stationary on either curve.
B, number of binodes of surface.
G, number of cnicnodes.

Article 9 (67.). And the accented letters have the like significations in regard to the reciprocal surface; referring them to the original surface, we have

n',	class of the surface.
a',	class of curve of intersection by any plane.
δ',	number of double tangents of curve of intersection.
κ',	number of its inflexions.
V,	class of node-couple torse.
k',	number of its apparent double planes.
i',	number of its triple planes.
q,	its order.
p,	order of node-couple curve.
y,	number of pinch-planes.
c',	class of spinode torse.
A',	number of its apparent double planes.
r',	its order.
σ',	order of spinode curve.
g',	number of off-planes.
γ',	number of close-planes.
θ',	number of common planes of node-couple and spinode torses.

Article 10 (68.). It is hardly necessary to recall that a spinode plane is a tangent plane meeting the surface in a curve having at the point of contact a spinode or cusp; the envelope of the spinode planes is the spinode torse, and the locus of their points of contact is the spinode curve. And similarly a node-couple plane is a tangent plane meeting the surface in a curve having two nodes; the envelope of the node-couple planes is the node-couple torse, and the locus of the points of contact the node-couple curve. The other terms made use of are all explained in the present Memoir.

Addition, August 3, 1869.

As in the theory of Curves, so in that of Surfaces, there are certain functions of the order, class, &c. and singularities which have the same values in the original and reciprocal figures respectively; for convenience I represent any such identity by means of the symbol $\mathfrak{A}$, *viz.*. $\phi(n, a, b, \ldots) = \mathfrak{A}$ denotes that the function $\phi(n, a, b, \ldots)$ is equal to the same function $\phi(n', a', b', \ldots)$ of the accented letters. By what precedes we have $a = \mathfrak{A}$ and it is moreover clear that any function of the unaccented letters which is $= 0$, or which is equal to a symmetrical function of any of the accented and unaccented letters, or to a function of a, is $= \mathfrak{A}$; for instance, from the equations of No. 45 we have $3n' - \kappa' = 3n - c$, and thence $3n - c - \kappa = 3a' - \kappa' - \kappa$, that is, $3n - c - \kappa = \mathfrak{A}$.

No. **5.** From one of the equations of No. 11 we have $n - 2C - 4B + \kappa - \sigma - 2j - 3\chi = n + n' - a = \mathfrak{A}$; we have thus the system of eight equations,

$$
\begin{aligned}
a &= \mathfrak{A}, \\
3n - c - \kappa &= \mathfrak{A}, \\
a(n-2) - \kappa + B - p - 2\sigma &= \mathfrak{A}, \\
b(n-2) - p - 2\theta - 3y - 3\iota &= \mathfrak{A}, \\
c(n-2) - 2\sigma - 4\rho - \gamma - 6\iota &= \mathfrak{A}, \\
2q - 2p + \beta + 3\iota + i &= \mathfrak{A}, \\
3r + c - \sigma a - \beta - 4\rho + 2\iota + \lambda &= \mathfrak{A}, \\
n - 2C - 4eB + \kappa - a - 2j - 3\chi - n + n' - a &= \mathfrak{A}.
\end{aligned}
$$

or if from these we eliminate κ, p, σ, ρ, then the system of five equations,

$$
\begin{aligned}
a &= \mathfrak{A}, \\
n(c-8) - 4\beta - \gamma - \lambda + 4\theta + 8\iota + 6\chi + 4j &= \mathfrak{A}, \\
(a-b)(n-2) - 11n + 3c + 4C + 9\iota + 2\rho + 3\gamma + 3\iota + 6\chi + 4j &= \mathfrak{A}, \\
3r - 20a + 6c - \beta + 2\iota + 10C + 20\iota + 16\rho + 10j - 4\theta &= \mathfrak{A}, \\
2q - 2b(n-2) + 5\beta + 6\gamma + 6\iota + 3\iota + j &= \mathfrak{A}.
\end{aligned}
$$

Dr. By means of a theorem of Clebsch's I was led to the following expression for the "deficiency" of a surface of the order n having the singularities considered in the foregoing Memoir:

$$\text{Deficiency} = \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-3)(b+c) + \tfrac{1}{2}(\delta+\kappa) + 2\iota + 4\rho + \tfrac{3}{2}\gamma + \tfrac{1}{2}\iota - \tfrac{1}{2}\theta.$$

This should be equal to the deficiency of the reciprocal surface, *viz.*. we must have

$$\tfrac{1}{2}(n-1)(n-2)(n-3) - 12(n-3)(b+c) + 6\delta + 6\kappa + 24\iota + 42\rho + 30\gamma + 12\iota - \tfrac{9}{2}\theta = \mathfrak{A};$$

but from a combination of the last-mentioned five equations we have

$$
\begin{aligned}
&-2n^2 - 6n^2 + 4n + (12n-36)b + (12n-48)c - 6\delta - 6\kappa - 24\iota \\
&\qquad - 42\rho - 30\gamma - 13\iota - 7j - 8\chi + 2\theta - 4C - 10\iota = \mathfrak{A};
\end{aligned}
$$

and adding to the last preceding equation we have

$$26n - 12c + \tfrac{1}{2}\gamma - \iota' - 7j - 8\chi + \tfrac{1}{2}\theta - 4C - 10\iota = \mathfrak{A}.$$

Substituting for $\mathfrak{A}$ its value in terms of the accented letters, we obtain for θ' the value

$$\theta' = 13' + 26n - 12c + \tfrac{1}{2}\gamma' + 7j' + 8\chi' + \tfrac{1}{2}\theta' + 4C' + 10\iota' - 26b' + 12c' - \iota - 7j - 8\chi + \tfrac{1}{2}\theta - 4C - 10\iota.$$

We have $c' = -3a + n + 3n'$.

$n = 2$. We have

$$26n - 12c + \tfrac{1}{2}\beta - \iota - 7j - 8\chi + \tfrac{1}{2}\theta - 4C - 10\iota = \mathfrak{A}$$

and multiplying by -4 and adding, the equation to be verified is

$$13n(c-8) - 13(4\beta + \gamma) + n + 4\iota' + 28j + 32\chi - 2\theta + 16C + 40\iota = \mathfrak{A}.$$

But we have from the Memoir

$$-13n(c-8) + 13(4\beta + \gamma) - 52j - 78\chi + 13\theta - 52C - 104\iota = \mathfrak{A},$$

which reduces the equation to

$$n + 4\iota - 24j - 46\chi + 11\theta - 36C - 64\iota = \mathfrak{A}$$

or substituting for 11θ its value, this is

$$(4\chi' - 4\chi - 336)\iota - (\tfrac{1}{2}g - 10)(16B + 8\chi + \theta) = \mathfrak{A},$$

an equation which is satisfied if

$$\chi' - \chi - 84 = 0, \quad \tfrac{1}{2}g - 10 = 0,$$

and

$$g = 20, \quad \text{or else} \quad 165 + 8\chi + \theta = 165' + 8\chi' + \theta'.$$

[412]

A Memoir on Cubic Surfaces.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLIX. (for the year 1869), pp. 231—326. Received November 12, 1868,—Read January 14, 1869.]

The present Memoir is in a measure supplementary to the Memoir "On the Distribution of Surfaces of the Third Order into Species, in reference to the presence or absence of Singular Points, and the reality of their Lines," by Professor Schläfli, Phil. Trans. vol. CLIII. (1863), pp. 193—241. But the object of the present Memoir is different; I attend to the several cases of cubic surfaces as grouped together in the Table, and referred to by the affixed roman numbers, for the purpose of obtaining the equations in point-coordinates of the several cases respectively, and the equations in plane-coordinates of the reciprocal surfaces. The memoir gives (partially or completely developed) the equations of the twenty-three cases depending on the nature of the singularities. It was written in connexion with these investigations on Cubic Surfaces, that the Memoir, [411], "Memoir on the Theory of Reciprocal Surfaces," was referred to; and the latter part of the Memoir is divided into sections headed thus:

[411], The twenty-three Cases of Cubic Surfaces.

&c. &c., referring to the several cases of the cubic surface, but the paragraphs are numbered continuously throughout the Memoir.

I prefer to make the ultimate division depending on the reality of the lines, presence or absence of singularities, or, what is the same thing, attending only to the reality, the division into (twenty-two, or as I reckon it) twenty-three cases.

Article Nos. 1 to 13. — Explanations and Table of Singularities.

Article 11 (1.). I designate as follows the twenty-three cases of cubic surfaces, adding to each of them its equation:

$$\begin{array}{rr}
\mathrm{I} = 12, & (X,Y,Z,W)^3 = 0, \\
\mathrm{II} = 12 - C_2, & W(a,b,c,f,g,h\overline{X,Y,Z})^2 + 2kXYZ = 0, \\
\mathrm{III} = 12 - A, & 2W(X+Y+Z)(lX+mY+nZ) + 2kXYZ = 0, \\
\mathrm{IV} = 12 - 2C_2, & WXZ + Y^2(yZ+\delta W) + (a,b,c,d\overline{X,Y})^3 = 0, \\
\mathrm{V} = 12 - B_3, & WXZ + (X+Z)(Y^2 - aX^2 - bZ^2) = 0, \\
\mathrm{VI} = 12 - B_3 - G_2, & WXZ + Y^2Z + (a,b,c,d\overline{X,Y})^3 = 0, \\
\mathrm{VII} = 12 - B_3, & WXZ + Y^2Z - Z^3 = 0, \\
\mathrm{VIII} = 12 - 3C_2, & WXZ + (a,b,c,d\overline{X,Y})^3 = 0, \\
\mathrm{IX} = 12 - 2B_3, & WXZ + (X+Z)(Y^2 - X^2) = 0, \\
\mathrm{X} = 12 - B_3 - G_2, & WXZ + Y^2Z + X^2 - Z^2 = 0, \\
\mathrm{XI} = 12 - B_3, & W(X+Y+Z)^2 + 2XYZ = 0, \\
\mathrm{XII} = 12 - B_3, & WXZ + Y^2(X+Y+Z) = 0.
\end{array}$$

Article 12 (2.)**.** Where C_2 denotes a conic-node diminishing the class by 2; B_3, B_4, B_5, B_6, a biplanar-node diminishing (as the case may be) the class by 3, 4, 5, or 6; U_6, U_7, U_8 a uniplanar-node diminishing (as the case may be) the class by 6, 7, or 8.

The affixed explanation, which I shall usually retain in connexion with the Roman number, shows therefore in each case what the class is, and also the singularities which cause the reduction: thus $\text{XIII} = 12 - B_3 - 2C_2$ indicates that there is a biplanar node, B_3, diminishing the class by 3, and two conic-nodes, C_2, each diminishing the class by 2; and thus that the class is $12-3-2\cdot 2 = 5$. As regards the cases XXII and XXIII, these are surfaces having a nodal right line, distinct from the nodal right line of a scroll; and are consequently scrolls, each having a simple directrix right line; the scroll $S(1,1)$ having a simple directrix right line coincident with the nodal line is the scroll $S(1,1)$; see my "Second Memoir on Skew Surfaces, otherwise Scrolls," Phil. Trans. vol. CLIV. (1864), pp. 559–577, [340].

Article 13 (3.)**.** The nature of the points $C(=C_2)$ and $B(=B_3)$ requires to be explained.

C $(=C_2)$ is a conic-node, where, instead of the tangent plane, we have a proper quadric cone.

B $(=B_3, B_4, B_5$ or $B_6)$ is a biplanar-node, where the quadric cone becomes a plane-pair (two distinct planes): the two planes are called the biplanes, and their line of intersection the edge. In B_3, the edge is not a line on the surface; in B_4, the tangent plane is touched along the edge by the surface, the edge is called the torsal edge; in B_5, the tangent plane coincides with one of the biplanes; in B_6, the tangent plane coinciding with one of the biplanes becomes oscular.

U $(=U_6, U_7$ or $U_8)$ is a uniplanar-node, where the quadric cone becomes a coincident plane-pair; say, the plane is the uniplane. It is to be observed that there is not in this case any edge. The uniplane meets the cubic surface in three lines, or say "rays," passing through the uniplanar-node, viz.

- In U_6, the rays are three distinct lines;
- In U_7, two of them coincide;
- In U_8, they all three coincide.

Article 14 (4.)**.** To connect these singular points with the theory of the preceding Memoir, it is to be observed that they are equivalent to a certain number of cnicnodes $C(=C_2)$ and binodes $B(=B_3)$ respectively; viz. we have

$$\begin{array}{ll} C = C, & B_3 = B, \\ B_4 = C + B, & B_5 = 2C + B, \\ B_6 = 3C, & U_6 = 3C, \\ U_7 = 2C + B, & U_8 = C + 2B. \end{array}$$

Article 15 (5.)**.** I take the opportunity of remarking that although the expressions cnicnode and binode properly refer to the simple singularities C and B, yet as $C_2 = C$, C_2 is properly spoken of as a cnicnode, and we may (using the term binode as an abbreviation for biplanar-node) speak of any of the singularities B_3, B_4, B_5, B_6 as a binode. Thus the surface $\text{X} = 12 - B_4 - C_2$ has a binode B_4 and a cnicnode C_2; although theoretically the binode B_4 is equivalent to two cnicnodes, and the surface belongs to those with three cnicnodes, or for which $C = 3$.

I use also the expression unode for shortness, instead of uniplanar-node, to denote any of the singularities U_6, U_7, U_8.

Article 16 (6.)**.** The foregoing equations (substantially the same as Schläfli's) are Canonical forms; the reduction of the equation of any case of surface to the above form is not always obvious. It would appear that each equation is from its simplicity in the form best adapted to the separate discussion of the surface to which it belongs; there is the disadvantage that the equations do not always (when from the geometrical connexion of the surfaces they ought to do so) lead the one to the other. This would be a serious imperfection if the equations were an end, the object in the present Memoir being only to establish the geometrical theory of the surfaces to which they respectively belong, and the imperfection is not material.

Article 17 (7.)**.** I have used the capital letters (X, Y, Z, W) in place of Schläfli's (x, y, z, w), reserving these for the plane-coordinates of the cubic surfaces, or (what is the same thing) point-coordinates of the reciprocal surfaces; and in the several cases I have interchanged the coordinates (in place of his (p, q, r, s) or (what is the same thing) so as to obtain a greater uniformity in the representation of the surfaces.

Article 18 (8.)**.** To explain this, let A, B, C, D be the vertices of the tetrahedron formed by the coordinate planes $A = YZW$, $B = ZWX$, $C = WXY$, $D = XYZ$; the coordinate planes have been chosen so that the vertices of the tetrahedron shall correspond to determinate singularities of the surface.

Consider first the surfaces which have no binodes B or unodes U. It is clear that the nodes might have been taken at any vertices whatever of the tetrahedron; they are taken thus: there is always a node C_2 at D; when there is a second node C_2, this is at C, the third one is at A, and the fourth at B.

Article 19 (9.)**.** Consider next the surfaces which have a binode B_3, B_4, B_5, or B_6; this is taken to be at D, and the biplanes to be $X = 0$, $Z = 0$ (the edge being therefore DB); viz. in B_3 or B_4, where the distinction arises, $X = 0$ is the ordinary biplane, $Z = 0$ the torsal or (as the case may be) oscular biplane. If there is a second node, it may be and is taken to be in the biplane $Z = 0$, at C, when the binode is B_3 or B_4, and it may be a node C_2. It is only when the binode is B_3 or B_4 that there can be a third node, for it is only in these cases that there is a second ordinary biplane $Z = 0$; but in these cases respectively the third node, a C_2, may be and is taken to be in the biplane $Z = 0$, at A.

Article 20 (10.)**.** The only case of two binodes is when each is a B_3. Here the first is at D, its biplanes being $X = 0$, $Z = 0$; and the second at C, the biplanes thereof are then $X = 0$ (which is the common biplane), and $W = 0$. If there is a third node, it may be and is taken to be at A; if a C_2, the biplanes of the second binode are $Z = 0$, $W = 0$; and the plane $Y = 0$ of the first binode is then a remaining biplane which will in either case be the plane through the three binodes D, C, A.

Article 21 (11.)**.** If there is a unode, this is taken to be at D, and its uniplane is $X + Y + Z = 0$. There is never, besides the unode, any other node.

Article 22 (12.)**.** The result is that the binodes, when there are such, are taken to be at D, C, A, in the order of their speciality, and that (except in the case XII $= 12 - U_8$) the biplanes are $X = 0$, $Z = 0$ (for the first binode, $Z = 0$ being the ordinary biplane, $Z = 0$ the special biplane), those of the second binode $Z = 0$, $W = 0$, those of the third binode $Z = 0$, $W = 0$, and that (except in the case XII $= 12 - U_8$) the uniplane is $Z = 0$. For example, in the surface XVII $= 12 - 2B_3 - C_2$, as represented by its equation $WXZ + Y^2Z + Z^2 = 0$, we have a B_3 at D, the biplanes being $Z = 0$, $Z = 0$, a B_3 at C, the biplanes being $Z = 0$, $W = 0$ (therefore $Z = 0$ the common biplane), and a C_2 at A.

In the case, however, of a single B_3, III $= 12 - B_3$, the biplanes are taken to be $X + Y + Z = 0$, $lX + mY + nZ = 0$.

Article 23 (13.)**.** It will be convenient (anticipating the results of the investigations contained in the present Memoir) to give at once the following Table of Singularities; the several symbols have of course the significations explained in the former Memoir.

	n	a	k	ρ	ι	B	C	n'	a'	δ'	κ'	b'	k'	t'	q	p	y	c'	r'	σ'
I	12	6	6	27	216	45	0	12	6	6	27	27	216	45	27	27	0	24	2	12
II	10	6	4	16	72	15	1	6	4	3	15	15	60	15	15	15	0	12	1	6
III	9	6	3	10	38	7	0	4	3	1	9	9	24	9	9	9	0	10	1	6
IV	8	6	2	6	17	3	2	3	3	0	7	7	12	7	7	7	0	10	1	6
V	8	6	2	6	16	2	1	4	3	1	7	7	12	7	7	7	0	8	0	4
VI	7	6	1	3	6	1	1	3	3	0	6	6	6	6	6	6	0	8	1	5
VII	7	6	1	3	4	0	0	6	4	3	6	6	6	6	6	6	0	6	0	3
VIII	6	6	0	1	1	0	0	6	4	3	3	3	3	1	3	3	1	9	2	6
IX	6	6	0	1	0	0	2	4	3	1	4	4	4	4	4	4	0	6	0	3
X	6	6	0	1	0	0	1	5	4	2	5	5	5	5	5	5	0	5	0	2
XI	6	6	0	0	0	0	0	6	5	3	6	6	6	6	6	6	0	6	0	3
XII	6	6	0	0	0	0	0	6	5	3	6	6	6	6	6	6	0	4	0	1
XIII	5	6	0	0	0	0	2	3	3	0	3	3	3	3	3	3	0	6	1	4
XIV	5	6	0	0	0	0	1	4	4	1	4	4	4	4	4	4	0	4	0	2
XV	5	6	0	0	0	0	0	5	5	2	5	5	5	5	5	5	0	5	0	2
XVI	4	6	0	0	0	0	3	3	3	0	3	3	3	3	3	3	0	4	0	2
XVII	4	6	0	0	0	0	1	5	5	2	5	5	5	5	5	5	0	4	0	2
XVIII	4	6	0	0	0	0	0	6	6	3	6	6	6	6	6	6	0	4	0	2
XIX	3	6	0	0	0	0	0	6	6	3	6	6	6	6	6	6	0	3	0	1
XX	3	6	0	0	0	0	0	6	6	3	6	6	6	6	6	6	0	3	0	1
XXI	3	6	0	0	0	0	0	9	9	6	9	9	9	9	9	9	0	3	0	1
XXII	3	3	0	0	0	0	0	3	3	1	3	1	0	0	3	1	0	3	1	3
XXIII	3	3	0	0	0	0	0	3	3	1	3	1	0	0	3	1	0	4	2	4

Continued Table.

	g'	γ'	θ'	i'	j'	χ'	B'	C'	β'	τ
I	16	8	16	6	2	4	0	0	54	0
II	8	4	8	3	1	2	0	0	30	0
III	4	2	4	1	0	1	0	0	18	0
IV	2	1	2	1	0	1	0	0	13	0
V	4	2	4	1	0	1	0	0	12	0
VI	2	1	2	1	0	1	0	0	8	0
VII	2	1	2	0	0	0	0	0	6	0
VIII	3	2	3	0	0	0	0	1	6	0
IX	2	1	2	0	0	0	0	0	4	0
X	1	0	1	0	0	0	0	0	3	0
XI	2	1	2	0	0	0	0	0	2	0
XII	1	0	1	0	0	0	0	0	1	0
XIII	2	1	2	0	0	0	0	0	3	0
XIV	1	0	1	0	0	0	0	0	2	0
XV	1	0	1	0	0	0	0	0	1	0
XVI	1	0	1	0	0	0	0	0	3	0
XVII	1	0	1	0	0	0	0	0	1	0
XVIII	1	0	1	0	0	0	0	0	0	0
XIX	0	0	0	0	0	0	0	0	0	0
XX	0	0	0	0	0	0	0	0	0	0
XXI	0	0	0	0	0	0	0	0	0	0
XXII	0	0	0	0	0	0	0	0	0	0
XXIII	0	0	0	0	0	0	0	0	0	0

Article Nos. 14 to 19. — Explanation in regard to the Determination of certain Singularities.

Article 24 (14.)**.** In the several cases I to XXI, we have a cubic surface ($n = 3$), with singular points C and B but without singular lines. The section by an arbitrary plane is thus a curve, order $n = 3$, that is, a cubic curve, without nodes or cusps, and therefore of the class $a' = 6$, having $\delta' = 6$ double tangents and $\kappa' = 9$ inflexions. The tangent cone with an arbitrary point as vertex is a cone of the order $a = 6$, having in the case I = 12 nodal lines and $\kappa = 6$ cuspidal lines, but with (in the several other cases) C nodal lines and B cuspidal lines (or rather singular lines tantamount to C double lines and B cuspidal lines): the class of the cone, or order of the reciprocal surface, is thus $n' = 6 \cdot 5 - 2(C + C) - 3(B + B) = 12 - 2C - 3B$.

Article 25 (15.)**.** In the general case I = 12, there are on the cubic surface 27 lines; these lie by 3's in 45 planes; the 27 lines constitute the node-couple curve of the order $p = 27$, and the node-couple torse consists of the pencils of planes through these lines respectively, being thus of the class $p = k' = 27$; the 45 planes are triple tangent planes of the node-couple torse, which has thus $t' = 45$ triple tangent planes. But in the other cases it is only certain of the 27 lines, say the "facultative lines" (as will be explained), which constitute the node-couple curve of the order p; the pencils of planes through these lines constitute the node-couple torse of the class $b' = p'$; the t' planes, each containing three facultative lines, are the triple tangent planes of the node-couple torse.

Or if we refer to the reciprocal surface, the numbers b', t' are then the order of the nodal curve, and the number of triple points of the nodal curve. Inasmuch as the nodal curve consists of right lines, the number k' of its apparent double points is given by the formula $2k' = b'^2 - b' - 3t'$, and comparing with the formula $q' = b'^2 - b' - 2k' - \frac{3}{2}y' - 6i'$ we have $q' + \frac{3}{2}y' = 0$, that is, $q' = 0$ (q the class of the nodal curve), and also $\gamma' = 0$.

Article 26 (16.)**.** In the general case I = 12, the spinode curve is the complete intersection of the cubic surface by the Hessian surface; it is thus of the order $\sigma' = 12$; but in the other cases the complete intersection consists of the spinode curve together with certain right lines not belonging to the curve, and the spinode curve is of an order less than 12: this will be further explained, and the reduction accounted for (see post, Nos. 24 *et seq.*).

Article 27 (17.)**.** Again, in the general case I = 12, each of the 27 lines is a double tangent of the spinode curve, and the tangent planes of the surface at the points of contact are the common planes of the spinode torse and the node-couple torse; or we have $\beta' = 2p' = 54$. In the other cases, however, we must take only the facultative lines, each of which is a double or single tangent of the spinode curve, and the tangent planes of the surface at the points of contact are the common tangent planes as above; that is, the number of contacts gives β' not in general $= 2p'$.

Article 28 (18.)**.** There are not, except as above, any common tangent planes of the two torses, so that not only $\gamma' = 0$, but also $i' = 0$. I do not at present account a priori for the values $\theta' = 16, 8$, and 16, which present themselves in the Table.

Article 29 (19.)**.** The cubic surface cannot have a plane of conic contact, and we have thus in every case $C' = 0$; but the value of B' is not in every case $= 0$. In what precedes we see how a priori the equation of the discussion of the reciprocal surface leads to the values of the equation of the nodal curve of the order b', and how the equation of the cuspidal curve might be obtained as the reciprocal of the spinode-torse of the cubic surface; but this is a laborious process, and it is in each case less necessary, inasmuch as the cuspidal curve is obtained by means of the equation of the reciprocal surface, putting in evidence the values $b', t', \beta', \gamma', i', j', \chi', B', C'$, known by means of the equation of the nodal curve.

Article Nos. 20 to 23. — The Lines and Planes of a Cubic Surface; Facultative Lines; Explanation of Diagrams.

Article 30 (20.)**.** In the general surface I = 12, we have 27 lines and 45 triple-tangent planes: through each line pass 5 planes, and in each plane lie 3 of the 27 lines. If we attend to the surfaces II to XXI (the scrolls XXII and XXIII do not come under the present considerations), in the several cases several of the lines coincide with each other; several of the planes also come to coincide with each other; but the sum of the frequencies is always the same. For each particular line, the line has a frequency (this represents the number of lines in the particular plane, the plane having a multiplicity, and the sum of these frequencies is $= 27 \times 5 = 135$.

Each plane has a multiplicity, and the frequency of each plane is the multiplicity of the line into the frequency of the line in regard to the plane.

Article 31 (21.)**.** The mode of coincidence of the lines and planes, and the several distinct lines and planes which are situate in or pass through the several distinct planes and lines respectively, are shown in the annexed diagrams I to XXI[(*)].

[(*)] *See the commencements of the several sections.*

Article 32 (22.)**.** For the explanation of the diagrams, I use the following terms: I call a line joining two nodes an "axis"; a line through a node not joining any other node is a "ray"; an edge of a binode is also a ray; any other line is a "mere line." An axis is always torsal or oscular; a ray may be torsal or oscular, but is not a mere line. The line joining a node to the point of contact of the plane touching along an axis is the "transversal" of such axis; in the case XVI $= 12 - 4C_2$, each transversal is a ray.

Article 33 (23.)**.** In the general case I = 12, each of the 27 lines is a facultative line, and is part of the node-couple curve; and the node-couple curve is, as already mentioned, made up of the 27 lines, thus a curve of the order 27. In fact each plane through a line meets the cubic surface in the line and in a conic; the line and conic meet in two points, and the plane (any plane) through the line is thus a double tangent plane touching the surface at the two points in question; the locus of the points of contact is the node-couple curve. But in the other cases, II to XXI, certain of the lines do not belong to the node-couple curve. To fix the ideas, consider the surface II $= 12 - C_2$: there are through the node six nodal rays, say rays 1, 2, 3, 4, 5, 6; attending to any one of these, a plane through the ray meets the surface in the ray itself and a conic; but the ray itself does not touch the surface at the node, and the plane through the ray is only a single tangent plane, not a double tangent plane; the ray is not part of the node-couple curve. We say that a line of the surface is or is not "facultative" according as it does or does not form part of the node-couple curve.

Article Nos. 24 to 26. — Axis; the different kinds thereof.

Article 34 (24.)**.** A line joining two nodes is an axis; such a line is always a torsal or oscular line, and it is not a mere line, but some further distinctions are requisite. Using the expressions in their strict sense, cnicnode $= C$, binode $= B$, an axis is a CC-axis, joining two cnicnodes, or it is a CB-axis joining a cnicnode and a binode, or it is a BB-axis joining two binodes. A CC-axis is torsal, the transversal being a mere line; a CB-axis is torsal, the transversal being a ray of the binode; a BB-axis is oscular.

The distinction is of course carried to the higher nodes B_4, B_5, B_6 and uniplanar nodes U_6, U_7, U_8; thus $(B_4 = B + C)$ the edge of a binode B_4 is not an axis at all, but $(B_4 = 2C)$ the edge of a binode B_4 is a CC-axis; $(B_5 = 2C + B)$ the edge of a binode B_5 is a twice-taken CC-axis; $(B_6 = 3C)$ the edge of a binode B_6 is a thrice-taken CC-axis; $(U_6 = 3C)$ each of the three rays is regarded as a CC-axis; $(U_7 = 2C + B)$ the double ray is regarded as a CC-axis + a twice-taken CB-axis; $(U_8 = C + 2B)$ the single ray is a CC-axis + a twice-taken CB-axis.

Article 35 (25.)**.** It has been mentioned that the intersection of the surface with the Hessian consists of the spinode curve together with certain right lines; these lines are in fact the axes. An examination of the several cases shows that in the complete intersection each CC-axis presents itself twice, each CB-axis 3 times, and each BB-axis 4 times. We thus see that a CC-axis, or rather the torsal plane along such axis, is the pinch-plane or singularity $y = 1$; the CB-axis, or rather the torsal plane along such axis, the close-plane $\gamma' = 1$; and the BB-axis, or oscular plane along such axis, the bitrope or singularity $\beta' = 1$; σ' the order of the spinode curve; for a cubic surface with singular lines the expression of a' being in fact $a' = 12 - 2j' - 3\gamma' - 4\beta'$.

Article 36 (26.)**.** I remark in further explanation, that in the several sections, in showing how the complete intersection of the cubic surface with the Hessian is made up, I have not referred to the axes in the above precise significations; thus for instance $\mathrm{XIV} = 12 - B_4 - C_2$, the binode B_4 is $C + B$, and the edge is thus a CB-axis, while the axis BC_2 is a CC-axis $(y' = 1 + 1 = 2,\ j' = 1)$. The complete intersection should therefore consist of the spinode curve, + edge (as a CB-axis) 3 times + axis (as a CC-axis) 2 times = 5 times; but without the full explanation it is stated in the section that the intersection is made up of the edge 4 times, and a cubic curve — which cubic curve together with the edge once constitutes the spinode curve; and so in other cases: this explanation will, I think, remove all difficulty.

Article Nos. 27 to 32. — On the Determination of the Reciprocal Equation.

Article 37 (27.)**.** Consider in general the cubic surface $(X, Y, Z, W)^3 = 0$, and in connexion therewith the equation $Xx+Yy+Zz+Ww = 0$, which regarding therein X, Y, Z, W as current coordinates, and x, y, z, w as constants, is the equation of a plane. If from the two equations we eliminate one of the coordinates, for instance W, we obtain

$$(Xw, Yw, Zw, -(Xx + Yy + Zz)^3 = 0,$$

which, (X, Y, Z) being current coordinates, is obviously the equation of the cone, vertex $(x = 0, y = 0, z = 0)$, which stands on the section of the cubic surface by the plane. Equating to zero the discriminant of this function in regard to (X, Y, Z), we express that the cone has a nodal line; that is, that the section has a node, or, what is the same thing, that the plane $xX + yY + zZ + wW = 0$ is a tangent plane of the cubic surface; and we thus by the process in fact obtain the equation of the reciprocal surface in plane-coordinates (x, y, z, w).

Article 38 (28.)**.** Consider in the same equation x, y, z, w as current coordinates, X, Y, Z, W as given parameters, the equation represents a system of three planes, viz. these are the planes $xX + yY + zZ + wW = 0$, where W has the three values given by the equation $(X, Y, Z, W)^3 = 0$; that is, the coordinates of any one of the three points of intersection of the line $x : y : z : w = \frac{1}{X} : \frac{1}{Y} : \frac{1}{Z} : -\frac{1}{W}$ with the cubic surface, and the equation is thus the equation of a system of 3 planes, the polar planes of three points of the cubic surface (which three points lie on an arbitrary line through the point $x = 0$, $y = 0$, $z = 0$).

In equating to zero the discriminant in regard to (X, Y, Z), we find the envelope of the system of three planes, or say of a plane, the polar plane of an arbitrary point on the cubic surface; this is, as is known, the same thing as the equation of the cubic surface in plane-coordinates (x, y, z, w).

Article 39 (29.)**.** In what precedes we have the explanation of a special process: the position of a point on the surface is determined by means of certain two parameters, the ratios $X : Y : Z$, or (as last explained) the position of the point is determined by the line joining this point with the point $(x = 0, y = 0, z = 0)$. More generally we may consider the position of the point as determined by means of any two parameters; the equation of the polar plane then contains the two parameters, and by taking the envelope in regard to the two parameters considered as variable, we have the equation of the reciprocal surface.

Article 40 (30.)**.** But let the parameters, say θ, ϕ, be regarded as varying successively; if we have on the surface a curve Θ, the equation whereof contains the parameter θ, and when θ varies this curve sweeps over the surface. The envelope in regard to ϕ of the polar plane of a point of the curve Θ may be a torse having its vertex on the reciprocal of the curve Θ; and the envelope of this torse by the variation of the parameter θ is the reciprocal surface. With a fixed line, the envelope of this plane is a sextic cone (the fixed line being $P = 0$, $Q = 0$), and by equating to zero the discriminant of a binary function of (P, Q) we obtain the equation of the sextic cone; assuming that the equation of the sextic cone has been obtained, this is the equation of the reciprocal surface.

Article 41 (31.)**.** In particular, if the section by any plane through the fixed line be a conic, its reciprocal is the quadricone, and the envelope of this quadricone is the reciprocal surface. This is a convenient one for obtaining the reciprocal of a cubic surface on which there are lines; what Schläfli does (but the process is not explained) in the several instances in which he obtains the equation of the reciprocal surface by means of a binary function. I remark that it would be very instructive, for each case of surface, to take the variable plane successively through the several kinds of lines on the particular surface; the equation of this quadricone would be thus obtained in different forms, putting in evidence the relation of the reciprocal surface to the fixed line.

Article 42 (32.)**.** It is to be mentioned that the equation of the reciprocal surface is found by equating to zero the discriminant of a ternary cubic, or a binary quartic, cubic, or quadric. The equation is in the form disc. = 0; contains a factor which for the adopted forms of equations is always a power or product of powers of w, z, x; known *a priori*, and which is thrown out without difficulty, the equation being thereby reduced to the proper order. There is the singular advantage that the process puts in evidence the existence of the cuspidal curve; and for a ternary cubic, the resulting equation would be simply $S^3 - 27T^2 = 0$, where $S = 0$, $T = 0$ are the equations of the cuspidal curve; but the factor thrown out as just mentioned occasions however a modification, viz. the intersection of the two surfaces is not an indecomposable curve, and the cuspidal curve is in most cases, not the complete intersection, but a partial intersection of the two surfaces.

In several cases it thus happens that the cuspidal curve is obtained as a complete intersection 3×4 or 2×6, without speciality. Similarly when the equation of the reciprocal surface is obtained by means of a binary cubic; if the coefficients hereof (functions of course of the coordinates x, y, z, w) be A, B, C, D, then the surface is

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0,$$

having the cuspidal curve $A, B, C = 0$, $B, C, D = 0$, subject however to modification in the case of a thrown out factor.

Article Nos. 33 and 34. — Explanation as to the Sections of the Memoir.

Article 43 (33.)**.** As regards the following Sections I to XXIII, it is to be observed that for the general surface I = 12, I do not attempt to form the equation of the reciprocal surface; and in some of the other cases, II = $12 - C_2$, &c., the equation of the reciprocal surface is not obtained in a completely developed form, or either the nodal curve or its cuspidal curve is not dealt with, so as to put in evidence the theory given in the former Memoir. Portions of the latter sections are consequently omitted; and in particular in the Section I there is given only the diagram of the 27 lines and the 45 planes (with however developments as to notation and otherwise which have no place in the subsequent sections), and the analytical expressions for the several lines and planes, although from the want of the equation of the reciprocal surface these have no present application. And so in some of the next following sections, no application is made of the analytical expressions of the lines and planes.

Article 44 (34.)**.** I call to mind that if a line be given as the intersection of the two planes

$$\begin{aligned} AX + BY + CZ + DW &= 0, \\ A'X + B'Y + C'Z + D'W &= 0, \end{aligned}$$

then the six coordinates of the line are

$$a, b, c, f, g, h = AD' - A'D,\ BD' - B'D,\ CD' - C'D,\ BC' - B'C,\ CA' - C'A,\ AB' - A'B,$$

and that in terms of the six coordinates the line is given as the common intersection of the four planes

$$(-h, g, a\,\overline{X, Y, Z, W})^2 = 0, \quad (-g, h, b\,\overline{X, Y, Z, W})^2 = 0, \quad \text{\&c.},$$

and that (reciprocating as usual in regard to $X^2 + Y^2 + Z^2 + W^2 = 0$) the coordinates of the reciprocal line are (f, g, h, a, b, c), that is, the common intersection of the four planes

$$(h, g, a\,\overline{x, y, z, w})^2 = 0, \quad (-g, h, -b\,\overline{x, y, z, w})^2 = 0, \quad \text{\&c.}$$

It is more convenient to consider a line as determined as the intersection of two planes rather than by means of its six coordinates; thus, for instance, to speak of the line $X = 0$, $Y = 0$ rather than of the line $(0, 0, 0, 1, 0, 0)$; and in some of the sections I have preferred not to give the expressions of the six coordinates of the several lines.

Article Nos. 35 to 46. — $\mathfrak{I} = 12$, Equation $(X, Y, Z, W)^3 = 0$.

35. There is in the system of the 27 lines and the 45 planes a complicated and many-sided symmetry which precludes the existence of any unique notation; the notation can only be obtained by starting from some arrangement which is not unique, but one of a system of several like arrangements. The notation employed in my original paper "On the Simple Tangent Planes of Surfaces of the Third Order," Camb. and Dub. Math. Journ. vol. IV. 1849, pp. 118–132, [76], and which is shown in the right hand and lower margins of the diagram, starts from such an arrangement; but it is so complicated that it can hardly be considered as at all putting in evidence the relations of the lines and planes; that of Dr Hart (Salmon, "On the Triple Tangent Planes of a Surface of the Third Order," same volume, pp. 252–260), depending on an arrangement of the 27 lines according to a cube of 3 each way, is a singularly elegant one, and will be presently reproduced.

But the most convenient one is Schläfli's, starting from a double-sixer; viz. we can (and that in 36 different ways) select out of the 27 lines two systems each of six lines, such that no two lines of the same system intersect, but that each line of the one system intersects all but the corresponding line of the other system; or, say, if the lines are

$$\begin{array}{cccccc} 1, & 2, & 3, & 4, & 5, & 6 \\ 1', & 2', & 3', & 4', & 5', & 6', \end{array}$$

then these have the thirty intersections

	1′	2′	3′	4′	5′	6′
1						
2						
3						
4						
5						
6						

Any two lines such as 1, $2'$ lie in a plane which may be called $12'$; similarly the lines $1'$, 2 lie in a plane which may be called $1'2$; these two planes meet in a line 12; and any three lines such as 12, 34, 56 meet in pairs, lying in a plane 12.34.56. We have thus the entire system of the 27 lines and 45 planes.

36. The diagram of the lines and planes is given in the original memoir.

37–39. The construction of double-sixers and trihedral-pairs is explained in detail in the original memoir.

38. It has been mentioned that the number of double-sixers was = 36; these are as follows:

$$\begin{array}{cccccc} 1, & 2, & 3, & 4, & 5, & 6 \\ 1', & 23, & 24, & 25, & 26 & \\ 2', & 13, & 14, & 15, & 16 & \\ & & \&c. & & & \end{array}$$

where, if we take any column of two lines, we have the complete number 216 pairs of non-intersecting lines.

39. We can out of the 45 planes select, and that in 120 ways, a *trihedral-pair*, that is, two triads of planes, such that the planes of the one triad, intersecting those of the other triad, give 9 lines. Analytically if $X = 0$, $Y = 0$, $Z = 0$ and $U = 0$, $V = 0$, $W = 0$ are the equations of the six planes, then the equation of the cubic surface is

$$XYZ + kUVW = 0.$$

See as to this post, No. 44.

The trihedral plane pairs are:

$$\begin{array}{ccc} 12', & 23', & 31' \\ 1'2, & 2'3, & 3'1 \\ & (\text{No. is} = 20) & \\ 12', & 34', & 14.23.56 \\ 2'3, & 4'1, & 12.34.56 \\ & (\text{No. is} = 90) & \\ 14.25.36, & 35.16.24, & 26.34.15 \\ 14.35.26, & 25.16.34, & 36.24.15 \\ & (\text{No. is} = 10) & \end{array}$$

Total $= 120$.

40. Dr Hart arranges the 27 lines, cubically, thus:

$$\begin{array}{ccc|ccc|ccc} A_1 & B_1 & C_1 & a_1 & b_1 & c_1 & \alpha_1 & \beta_1 & \gamma_1 \\ A_2 & B_2 & C_2 & a_2 & b_2 & c_2 & \alpha_2 & \beta_2 & \gamma_2 \\ A_3 & B_3 & C_3 & a_3 & b_3 & c_3 & \alpha_3 & \beta_3 & \gamma_3 \end{array}$$

where letters of the same alphabet denote lines in the same plane, if only the letters and suffixes are the same; thus A_1, A_2, A_3 lie in a plane $A_1A_2A_3$; letters of different alphabets denote lines which meet according to the Table.

41. I find that one way in which this may be identified with the double-sixer notation is to represent the above arrangement by

$$\begin{array}{cccccc} 1, & 2, & 3, & 4, & 5, & 6 \\ 1', & 2', & 3', & 4', & 5', & 6' \\ 12, & 13, & 14, & 15, & 16, & \ldots \end{array}$$

and then we may permute the i, j, k, l, m, n in 720 ways, and then upon the arrangements so obtained make any of the substitutions which permute inter se the 36 double-sixers.

42. The equations of the 45 planes are obtained taking the equation of the surface to be

$$W(1, 1, 1, 1, mn + \tfrac{1}{l}, nl + \tfrac{1}{m}, lm + \tfrac{1}{n}, l + \tfrac{1}{l}, m + \tfrac{1}{m}, n + \tfrac{1}{n}\overline{X, Y, Z, W})^2 + kXYZ = 0,$$

where $k = \frac{2(p-a)}{lmn}$, $a = lmn + \frac{1}{lmn}$, $p = lmn + \ldots$

43. The coordinates of the 27 lines are then found to be as given in the original memoir.

44. We have $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ for $(12' = w)$, $(23' = x)$, $(31' = \theta)$, $(12.34.56 = u)$; and representing by $(= lX + \frac{1}{l}Y + \frac{1}{n}Z + W)$ the equations of the planes $(14' = \xi)$, $(42' = y)$, $(21' = \eta)$, the equation of the cubic surface may be presented in the several forms

$$\begin{aligned} &= Fgg + k\eta ZX, \\ &= Fhh + k\xi XY, \\ &= W\xi\xi' + k\eta XY, \\ &= W\eta\eta' + k\xi ZX, \\ &= W\zeta\zeta' + k\xi XY, \\ &= Wpp' + k\xi YZ, \\ &= Wqq' + k\eta ZX, \\ &= Wrr' + k\xi XY, \quad \&\text{c.}, \end{aligned}$$

which are the 16 forms containing W, out of the complete system of 120 trihedral-pair forms.

45. The 27 lines are each of them a facultative line; each of the 27 lines is a double tangent of the spinode curve; therefore $b' = p' = 27$; $t' = 45$.

46. The nodal curve is composed of the 27 lines ($b' = 27$, $t' = 45$ ut supra); the spinode curve is a complete intersection 3×4; $\sigma' = 12$.

The equation of the reciprocal surface is not here investigated; its form is $S^3 - 27T^2 = 0$, where $S = 0$, $T = 0$ are the equations of the cuspidal curve; but for the factor thrown out as just mentioned, we should have simply $(S = 0, T = 0)$, or, as the case may be, $(S = 0, T = 0)$ for the existence of the cuspidal curve. The equation shows that the cuspidal curve is a complete intersection 6×4: $c' = 24$.

Section II $= 12 - C_2$. Article Nos. 47 to 59. Equation $W(a,b,c,f,g,h\overline{X,Y,Z})^2 + 2kXYZ = 0$.

47. It may be remarked that the system of lines and planes is at once deduced from that belonging to $\mathfrak{I} = 12$, by supposing that in the double-sixer the corresponding lines 1 and $1'$, &c. severally coincide; the line 12, instead of being given as the intersection of the planes $12'$, $1'2$, is given as the third line in the plane 12, which in fact represents the coincident planes $12'$ and $1'2$.

48. The diagram of the lines and planes is given in the original memoir. The lines are: 6 rays through the conic-node, and 15 mere lines. The planes are: 15 biradial planes (each containing a pair of rays), and 15 planes each containing three mere lines.

49. Putting the equation of the surface in the form

$$W(1,1,1,1+\ldots\overline{X,Y,Z})^2 + XYZ = 0,$$

where for shortness

$$\alpha = mn - l, \quad \beta = nl - m, \quad \gamma = lm - n, \quad \delta = lmn - 1, \quad p = lmn,$$

then taking $X = 0$ as the equation of the plane [12], $Y = 0$ as that of the plane [34], $Z = 0$ as that of the plane [56], the equations of the 30 distinct planes are found to be:

$$\begin{array}{lll} Z = 0, & [12] & mX + l^{-1}Y + Z = 0, \\ Y = 0, & [34] & m^{-1}X + l^{-1}Y + Z = 0, \\ Z = 0, & [56] & X + nY + mZ = 0, \\ & [23] & X + n^{-1}Y + mZ = 0, \\ & [14] & X + nY + m^{-1}Z = 0, \quad \text{\&c.} \end{array}$$

50. And the coordinates of the 21 distinct lines are given in the original memoir.

51. The six nodal rays are not, the fifteen mere lines are, facultative; hence $b' = p' = 15$; $t' = 15$.

52. Resuming the equation $W(a,b,c,f,g,h\overline{X,Y,Z})^2 + 2kXYZ = 0$, the equation of the Hessian surface is found to be

$$\begin{aligned} k^2W^2&(a,b,c,f,g,h\overline{X,Y,Z})^2 \\ &+ 2kW\{(a,b,c,f,g,h\overline{X,Y,Z})^2(FX+GY+HZ) - kXYZ\} \\ &- k\{a^2X^4 + b^2Y^4 + c^2Z^4 - 2bcY^2Z^2 - 2caZ^2X^2 - 2abX^2Y^2 \\ &\qquad - 4XYZ[(af+gh)X + (bg+hf)Y + (ch+fg)Z]\} = 0, \end{aligned}$$

where $(A,B,C,F,G,H) = (bc-f^2, ca-g^2, ab-h^2, gh-af, hf-bg, fg-ch)$, $K = abc - af^2 - bg^2 - ch^2 + 2fgh$.

The Hessian and cubic surfaces intersect in a complete intersection, which is the spinode curve; that is, the spinode curve is a complete intersection 3×4; $\sigma' = 12$.

The equations of the spinode curve may be written in the simplified form

$$\begin{aligned} W(a,b,c,f,g,h&\overline{X,Y,Z})^2 + 2kXYZ = 0, \\ &- 8KXYZW + 8kXYZ(afX + bgY + chZ) \\ &\quad - k^2\{a^2X^4 + b^2Y^4 + c^2Z^4 - 2bcY^2Z^2 - 2caZ^2X^2 - 2abX^2Y^2\} = 0, \end{aligned}$$

and it appears hereby that the node is a sixfold point on the curve, the tangents of the curve in fact coinciding with the six rays.

Each of the 15 lines touches the spinode curve twice; in fact, for the line 12 we have $X = 0$, $W = 0$: and substituting in the equations of the spinode curve, we have $(bY^2 - cZ^2)^2 = 0$; that is, we have the two points of contact $X = 0$, $W = 0$, $Y\sqrt{b} = \pm Z\sqrt{c}$. Hence $\beta' = 30$.

Reciprocal Surface.

53. The equation is found by equating to zero the discriminant of the ternary cubic function

$$(Xx + Yy + Zz)(a,b,c,f,g,h\overline{X,Y,Z})^2 - 2kwXYZ,$$

viz. the discriminant contains the factor w^2 which is to be thrown out, thus reducing the order to $n' = 10$.

54. Write as before (A, B, C, F, G, H) for the inverse coefficients, $K = abc - af^2 - bg^2 - ch^2 + 2fgh$; and moreover

$$\begin{aligned}
\Phi &= (A, B, C, F, G, H\overline{)x, y, z})^2, \\
P &= Ax + Hy + Gz, \quad Q = Hx + By + Fz, \quad R = Gx + Fy + Cz, \\
t &= fx + gy + hz, \\
U &= afyz + bgzx + chxy, \\
V &= 2Kxyz - aPyz - bQzx - cRxy \\
&= -aHy^2z - bFz^2x - cGx^2y - aGyz^2 - bHzx^2 - cFxy^2 \\
&\quad + (-abc - af^2 - bg^2 - ch^2 + 2fgh)xyz, \\
W &= (A, B, C, F, G, H\overline{)ayz, bzx, cxy})^2, \\
L &= kw^2t^2 - 2ktw - \Phi, \\
M &= kwU + V, \\
N &= 2kabc\,xyzw + W.
\end{aligned}$$

54. (continued) Then the invariants of the ternary cubic are

$$S = L^2 - 12kwM, \quad T = L^3 - 18kwLM - 54k^2w^2N,$$

and the required equation of the reciprocal surface is

$$S^3 - 27T^2 = 0,$$

viz. this is

$$(L^2 - 12kwM)^3 - 27(L^3 - 18kwLM - 54k^2w^2N)^2 = 0.$$

55. I present the result as follows; the coefficients deducible by mere cyclical permutations of the letters a, b, c and f, g, h, are indicated by (α).

$$\begin{aligned}
&(kw)^0 \cdot 2abc\,xyz \\
&\qquad + (kw)^1 \cdot [\ldots] \\
&\qquad + (kw)^2 \cdot [\ldots] \\
&\qquad + (kw)^3 \cdot [\ldots] \\
&\qquad - K\{(A, B, C, F, G, H\overline{)x, y, z})^2(cy^2 - 2fyz + bz^2)(az^2 - 2gzx + cx^2)(bx^2 - 2hxy + ay^2)\}.
\end{aligned}$$

56–59. In explanation of the discussion of the reciprocal surface:

Node C_2: $Z = 0$, $Y = 0$, $Z = 0$.

Reciprocal plane: $w = 0$.

Conic of contact: $(A, B, C, F, G, H\overline{)x, y, z})^2 = 0$, $w = 0$.

Nodal rays are sections of cone by planes $X = 0$, $bY^2 + 2fYZ + cZ^2 = 0$; $Y = 0$, $cZ^2 + 2gZX + aX^2 = 0$; $Z = 0$, $aX^2 + 2hXY + bY^2 = 0$.

Tangent cone is $(a, b, c, f, g, h\overline{)X, Y, Z})^2 = 0$, $w = 0$.

The equation shows that the nodal curve is made up of the conic $(A, B, C, F, G, H\overline{)x, y, z})^2 = 0$, twice, and of the six lines, tangents to this conic, which are the reciprocals of the nodal rays; these six lines are thus mere scrolar lines on the reciprocal surface.

As regards the cuspidal curve, the equation of the surface may be written

$$(S^3 - 12kwM)(4M^3 + S^2N) - (LM + 9kwN)^2 = 3\{SM^3 + S^3N - 18kwLMN - 16kwM^3 - 27k^2w^2N^2\} = 0,$$

and we thus have

$$4M^3 + S^2N = 0, \quad LM + 9kwN = 0, \quad S - 12kwM = 0,$$

or, what is the same thing,

$$\frac{L}{12M} = \frac{M}{-9N} = \frac{-9N}{L^2},$$

(equivalent to two equations) for the cuspidal curve. Attending to the second and third equations, the cuspidal curve may be considered as the residual intersection of the quartic and quintic surfaces $L^2 - 12kwM = 0$, $LM + 9kwN = 0$, which partially intersect in the conic $w = 0$, $\Phi = 0$; it is a curve $4 \cdot 5 - 2$; $c' = 18$.

Section III $= 12 - B_3$**. Article Nos. 60 to 72. Equation** $2W(X+Y+Z)(lX+mY+nZ)+2kXYZ=0$**.**

60. The system of lines and planes is at once deduced from that belonging to $\mathfrak{I}=12$, by supposing the tangent cone to reduce itself to the pair of biplanes; 3 of the planes (α) of $\mathfrak{II}=12-C_2$ thus coming to coincide with the one biplane, and three of them with the other biplane.

61. The diagram is given in the original memoir. The lines are: rays 1, 2, 3 in one biplane; rays 4, 5, 6 in the other biplane; 2 mere lines. The planes are: 2 biplanes; 9 biradial planes; 6 planes each containing three mere lines.

62. Taking $lX+mY+nZ=0$ for the biplane that contains the rays 1, 2, 3, and $X+Y+Z=0$ for that which contains the rays 4, 5, 6, we may take $X=0$, $Y=0$, $Z=0$ for the equations of the planes [14], [25], [36] respectively; and then writing for shortness

$$m-n=\lambda,\quad n-l=\mu,\quad l-m=\nu,$$

and assuming, as we may do, $k=\lambda\mu\nu$, so that the equation of the surface is

$$W(X+Y+Z)(lX+mY+nZ)+(m-n)(n-l)(l-m)XYZ=0,$$

the equations of the 17 distinct planes are found to be:

$$\begin{array}{ll} Z=0, & [14] \\ Y=0, & [25] \\ Z=0, & [36] \\ X+Y+Z=0, & [123] \\ lX+mY+nZ=0, & [456] \\ lX+nY+nZ=0, & [16] \\ \text{\&c.} & \end{array}$$

63. And the coordinates of the fifteen distinct lines are given in the original memoir.

64. The rays are not, the mere lines are, facultative; hence $b'=p'=9$; $t'=6$.

65. The equation of the Hessian surface is

$$\begin{aligned} &-W(X+Y+Z)(lX+mY+nZ)(\mu\nu X+\nu\lambda Y+\lambda\mu Z) \\ &\qquad -k(l^2X^4+m^2Y^4+n^2Z^4-2mnY^2Z^2-2nlZ^2X^2-2lmX^2Y^2) \\ &+kXYZ\{(l^2+3lm+3ln+mn)X+(m^2+3mn+3ml+nl)Y+(n^2+3nl+3nm+lm)Z\}=0. \end{aligned}$$

The Hessian and cubic surfaces intersect in an indecomposable curve, which is the spinode curve; that is, the spinode curve is a complete intersection 3×4; $\sigma'=12$.

The equations may be written in the simplified form

$$\begin{aligned} &W(X+Y+Z)(lX+mY+nZ)+kXYZ=0, \\ &l^2X^4+m^2Y^4+n^2Z^4-2mnY^2Z^2-2nlZ^2X^2-2lmX^2Y^2 \\ &\quad -4XYZ\{l(m+n)X+m(n+l)Y+n(l+m)Z\}=0. \end{aligned}$$

We may also obtain the equation

$$\begin{aligned} &k^2(X+Y+Z)(lX+mY+nZ)\{lX^2+mY^2+nZ^2-(m+n)YZ-(n+l)ZX-(l+m)XY\} \\ &\qquad +\lambda^2Y^2Z^2+\mu^2Z^2X^2+\nu^2X^2Y^2-2XYZ(\mu\nu X+\nu\lambda Y+\lambda\mu Z)=0, \end{aligned}$$

which shows that there is at B_3 an eightfold point, the tangents being given by $(X+Y+Z)(lX+mY+nZ)=0$.

Each of the facultative lines is a double tangent of the spinode curve; whence $\beta'=18$.

Reciprocal Surface.

66. The equation may be deduced from that for $\mathfrak{II}=12-C_2$; viz. writing

$$(a,b,c,f,g,h\overline{)X,Y,Z\Large)}^2=2(X+Y+Z)(lX+mY+nZ),$$

so that $(a,b,c,f,g,h)=(2l,2m,2n,m+n,n+l,l+m)$, we have

$$(A,B,C,F,G,H)=-(\lambda^2,\mu^2,\nu^2,\mu\nu,\nu\lambda,\lambda\mu);\quad K=0.$$

Writing also $\lambda,\mu,\nu=m-n,n-l,l-m$ as before,

$$\begin{aligned}\lambda x+\mu y+\nu z&=\sigma,\\ lmn\,xyz&=\theta,\\ l(m+n)yz+m(n+l)zx+n(l+m)xy&=\psi,\\ t\,xyz+\sigma\psi&=\phi,\end{aligned}$$

we have $U=2\psi$, $V=2\sigma\phi$, $W=-4\Phi$, and then

$$\begin{aligned}L&=k(\psi v^2-2ktw+\sigma^2),\\ M&=2(kw\psi+\sigma\phi),\\ N&=4(lmnkxyzw-\psi^2).\end{aligned}$$

67. The equation is

$$\begin{aligned}(k^2\psi^2-2kt\psi+\sigma^2)^2\{kW\theta+kw(-2t\theta+\psi^2-\phi^2)+\sigma^2\theta+2\sigma\phi\psi+2t\phi^2\}&\\ -32(k\psi\psi+\sigma\phi)^3-108k^2w^2(4\psi\theta-\psi^2)^2&=0.\end{aligned}$$

68. I verify the last term in the particular case $z=0$ as follows: the coefficient of σ^4 is

$$(\theta,2n(l+m)xy,2(m+n)x+2(n+l)y,\sigma^2\theta+\phi^2,n\nu xy)^2$$

which reduces to the general value $-4\lambda\mu\nu(y-z)(z-x)(x-y)(ny-mz)(lz-nx)(mx-ly)$.

69. In the discussion of the equation it is convenient to write down the relations of the two surfaces:

Cubic surface.	*Reciprocal surface.*
Plane $w=0$,	Plane $w=0$,
B_3	B_3'
Biplanes $X+Y+Z=0$, $lX+mY+nZ=0$.	
Rays in first biplane,	Points in $w=0$,
$Y+Z=0$; $Z+X=0$;	$y-z=0$, $z-x=0$, $x-y=0$;
$X+Y=0$	through first point, viz.
	$ny-mz=0$, $nz-lx=0$, $lx-my=0$.

70. The equation puts in evidence the line $\sigma=0$ (reciprocal of the edge) three times, and the six lines (reciprocals of the rays) each once. Observe that the edge is not a line on the reciprocal surface; and that the six lines (reciprocals of the rays) are mere scrolar lines on the reciprocal surface; they pass, three of them, through the point $x=y=z$, and the other three through the point $x:y:z=l:m:n$; that is, they are six tangents of the point-pair (reciprocal of the pair of biplanes) formed by these two points.

71. I do not attempt to put in evidence the nodal curve on the surface; by what precedes it consists of 9 lines, reciprocals of the mere lines. If we denote by 1, 2, 3 and 4, 5, 6 the lines which pass through the points $x=0$, $y=0$, $z=0$ and through the point $x:y:z=l:m:n$ respectively, then these intersect in the nine points 14, 15, 16, 24, 25, 26, 34, 35, 36; and through each of these there passes a nodal line which may be represented by the same symbol; that is, we have the nodal lines 14, ..., 36. Two lines such as 14, 25 meet; and three lines such as 14, 25, 36 meet in a point; we have thus the six points 14.25.36 &c. triple points on the nodal curve; as before, $b'=9$, $t'=6$.

72. The cuspidal curve is given by the equations

$$\begin{aligned}(k\psi v^2-2kt\psi+\sigma^2)^2-24k\psi(k\psi v+\sigma\phi)^2&=0,\\ (k\psi v^2-2kt\psi+\sigma^2)(k\psi v+\sigma\phi)+18\psi(lmnkxyzw-\psi^2)&=0.\end{aligned}$$

Writing down the two equations, these are respectively of the orders 4 and 5; but they intersect in the line $w=0$, taken four times, or say, the cuspidal curve is a partial intersection $4\cdot5-4$; $\sigma=4$; $c'=16$.

Section IV $= 12 - 2C_2$. **Article Nos. 73 to 84. Equation** $WXZ + Y^2(\gamma Z + \delta W) + (a, b, c, d\overline{X, Y})^3 = 0$.

73. The diagram of the lines is given in the original memoir. The lines are: axis [0] joining the two nodes; 4 rays through each node; 8 mere lines. The planes are: plane touching along axis; 16 planes through axis (each containing a ray of one node and a ray of the other); 24 biradial planes (each containing two rays of one node or two rays of the other); 3 planes each containing three mere lines.

74. Writing

$$X(a, b, c, d\overline{X, Y})^2 - \gamma\delta Y^2 = -\gamma\delta(f_1X - Y)(f_2X - Y)(f_3X - Y)(f_4X - Y),$$

the 20 planes are:

$$\begin{array}{ll} Z = 0, & [0] \\ X - f_1Y = 0, & [11'] \\ X - f_2Y = 0, & [22'] \\ X - f_3Y = 0, & [33'] \\ X - f_4Y = 0, & [44'] \\ \delta\{Z - (f_1 + f_2)Y\} - f_1f_2\gamma W = 0, & [12] \\ \&\text{c.} & \end{array}$$

75. And the 16 lines are given in the original memoir.

76. To verify the equations of the line $12.3'4'$, observe that the two equations give

$$\gamma Z + \delta W = \gamma\delta\{Z(f_1 + f_4) - Y(f_1f_4 + f_2f_3)\},$$

and substituting in the equation of the surface, this becomes an identity.

77. The facultative lines are the transversal and the six mere lines; $b' = p' = 7$.

78. The equation of the Hessian surface is found to be

$$\begin{aligned} (\gamma^2 + \delta^2W)XZW + Y^2(\gamma Z - \delta W)^2 + 3(cX + dY)XZW + 12\gamma\delta XY^2(aX + bY) \\ - (\gamma Z + \delta W)(3aZ^2 + 9bX^2 - Y^2 + 6cXY) \\ - 9X^2\{(ac - b^2)X^2 + (ad - bc)XY + (bd - c^2)Y^2\} = 0. \end{aligned}$$

79. Combining with the foregoing the equation of the surface

$$XZW + Y^2(\gamma Z + \delta W) + (a, b, c, d\overline{X, Y})^3 = 0,$$

it appears that $X = 0$, or $Y = 0$, and that the two equations have along the line $X = 0$, $Y = 0$ (the axis) the same tangent plane $X = 0$; they meet in the line $X = 0$, $Y = 0$ twice, and in a residual curve of the tenth order, which is the spinode curve.

The spinode curve may be presented in the somewhat more simple form

$$\begin{aligned} XZW + Y^2(\gamma Z + \delta W) + (a, b, c, d\overline{X, Y})^3 = 0, \\ - 4\gamma\delta Y^2ZW - 4(\gamma Z + \delta W)(a, b, c, d\overline{Z, Y})^3 + 12\gamma\delta XY^2(aX + bY) \\ + X^2(-12ac + 9b^2) - 3d(4aZ^2Y + 6bX^2Y^2 + 4cXY^3 + dY^4) = 0, \end{aligned}$$

the line $X = 0$, $Y = 0$ twice.

The spinode curve is of the tenth order; that is, we have $\sigma' = 10$.

Each of the 6 mere lines is a double tangent to the spinode curve; the transversal is only a single tangent: to show this, observe that the equations of the spinode curve contain the line $X = 0$, $Y = 0$ twice. For the transversal $X = 0$, $Y = 0$, $\gamma Z + \delta W + dY = 0$; substituting in the equations of the curve, the cubic surface equation is of course satisfied identically; for the second equation, writing $Z = 0$, this becomes $Y^2\gamma Z - \delta W = 0$. The value $Y^2 = \delta W/\gamma Z$ gives a point of contact not belonging to the spinode curve. Hence the number of contacts is $2 \cdot 6 + 1 = 13$; that is, we have $\beta' = 13$.

Reciprocal Surface.

80. The equation is found by equating to zero the discriminant of the binary quartic

$$\{xX^2 + yXY - (\delta z + \gamma w)Y^2\}^2 + 4Zw\{X(a, b, c, d\overline{Z, Y})^2 - \gamma\delta Y^2\},$$

or say this is $(X, Y)^4$, where the coefficients are

$$\begin{aligned}
&6a(x^2 + 24azw),\\
&3xy + 18bzw,\\
&y^2 - 2(\delta z + \gamma w)x + 12czw,\\
&-3(\delta z + \gamma w)y + 6dzw,\\
&6(\delta z - \gamma w).
\end{aligned}$$

81. Forming the invariants, these are

$$\begin{aligned}
\tfrac{1}{12}I &= A = \ldots + 24azw,\\
-\tfrac{1}{6}J &= A' + 36AUzw + 216Vz^2w^2 + 864Wz^3w^3,
\end{aligned}$$

where U, V, W are given in the original memoir.

82. It is convenient to modify the form of the equation as follows; write

$$U_1 = U + 8a\gamma\delta zw, \quad V_1 = V + (-8ac + 9b^2)\gamma\delta zw,$$

so that

$$\begin{aligned}
\Lambda &= y^2 + 4(\delta z + \gamma w)x,\\
U_1 &= -2\gamma\delta x^2 + 2a(\delta z + \gamma w)^2 + 3by(\delta z + \gamma w) + c[y^2 - 2(\delta z + \gamma w)x] - dxy,\\
V_1 &= (-8ac + 9b^2)(\delta z + \gamma w)^2\\
&\quad + (2c^2 - bd)[y^2 - 2(\delta z + \gamma w)x]\\
&\quad + (-4ad + 6bc)y(\delta z + \gamma w)\\
&\quad - 2cdxy + d^2x^2 + 4\gamma\delta(2cx^2 - 3bxy + ay^2),\\
\mu &= c^2 - bd,\\
\nu &= ad^2 - 2bcd + 2c^3,
\end{aligned}$$

Λ, U_1, V_1 being, it will be observed, functions of x, y, $\delta z + \gamma w$. The transformed equation is

$$\Lambda^2(\Lambda^2\mu - \Lambda V_1 + U_1^2) + \Omega zw = 0,$$

where the term Ω may be calculated without difficulty: the first term of this is

$$= [y^2 + 4(\delta z + \gamma w)x]^2 \cdot 4\gamma^2\delta^2[x + f_1 y - f_1^2(\delta z + \gamma w)] \ldots [x + f_4 y - f_4^2(\delta z + \gamma w)],$$

the developed expressions of $\frac{1}{4}(\Lambda^2\mu - \Lambda V_1 + U_1^2)$ and of $\gamma^2\delta^2$ into the product of the linear factors being in fact each

$$\begin{aligned}
= x^4 \cdot \gamma^2\delta^2 + x^3y \cdot d\gamma\delta + x^2y^2 \cdot (-3c\gamma\delta) + xy^3 \cdot 3b\gamma\delta + y^4 \cdot (-a\gamma\delta)&\\
+ [x^3(-d^2 - 6c\gamma\delta) + x^2y(3cd + 9b\gamma\delta) + xy^2(-3bd - 4a\gamma\delta) + y^3 \cdot ad](\delta z + \gamma w)&\\
+ [x^2(9c^2 - 6bd - 2a\gamma) + xy(3ad - 9bc) + y^2 \cdot 3ac](\delta z + \gamma w)^2&\\
+ [x(6ac - 9b^2) + y \cdot 3ab](\delta z + \gamma w)^3&\\
+ a^2\delta^4 \cdot (\delta z + \gamma w)^4.&
\end{aligned}$$

The form puts in evidence the section by the plane $w = 0$, which is the reciprocal of the node D, viz. this is a conic (the reciprocal of the tangent cone) twice, and four lines, the reciprocals of the nodal rays, each once. And similarly for the section by the plane $z = 0$.

83. The nodal curve is made up of the lines which are the reciprocals of the six mere lines and the transversal; viz. we have three pairs of lines and a seventh line, the lines of each pair intersecting at a point of the seventh line, and these three points being the triple points of the nodal curve; $t' = 3$ as before.

84. The equations of the cuspidal curve are at once reduced to the form

$$\Lambda^2 + 24Uzw + 144\mu z^2 w^2 = 0,$$
$$\Lambda U + (18V - 12\mu\Lambda)zw + 72\nu z^2 w^2 = 0,$$

which are two quartic surfaces having in common the conics $z = 0$, $\Lambda = 0$, and $w = 0$, $\Lambda = 0$; or we may say that the cuspidal curve is a curve $4 \cdot 4 - 2 - 2$; that is $c' = 12$.

This completes the discussion of the surface IV $= 12 - 2C_2$. The memoir continues with the remaining cases V to XXIII, which are discussed in subsequent sections with similar detail.